

- 2 The viscosity μ of a fluid can be determined by measuring the terminal velocity v_t of a sphere when it descends in the fluid. The fluid has a density ρ_f while the sphere has a density ρ_s and a diameter d . The viscosity can then be calculated using the equation

$$\mu = \frac{5(\rho_s - \rho_f)}{9v_t} d^2$$

The quantities measured are

$$v_t = (1.60 \pm 0.04) \text{ m s}^{-1}$$

$$\rho_s = (2700 \pm 20) \text{ kg m}^{-3}$$

$$\rho_f = (900 \pm 10) \text{ kg m}^{-3}$$

$$d = (20.0 \pm 0.4) \text{ mm}$$

What is the percentage uncertainty in the value of μ ?

A 6.2 %

B 7.1 %

C 8.2 %

D 30 %

Ans: C

$$\frac{\Delta\mu}{\mu} = \frac{30}{1800} + \frac{0.04}{1.60} + \frac{0.8}{20.0}$$

$$\frac{\Delta\mu}{\mu} \times 100\% = 8.2\%$$

